

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

# Chapter 3: A Story of Numbers

Topic-wise short notes with solved examples

How people learned to count and write numbers: tally marks, grouping, landmark numbers, the idea of a base, place value and zero, ending with the Hindu (Indian) number system used across the world today.

## 1 Counting and number systems

- To count a collection we match its objects one by one with a fixed, ordered sequence. This matching is a **one-to-one mapping**: no two objects are matched with the same item.
- The fixed sequence can be made of **objects** (sticks, pebbles), **names** (spoken words) or **written symbols**. Such a standard sequence is called a **number system**.
- The symbols of a written number system are called **numerals**.
- A good number system must never run out (numbers have no end) and must still be easy to use.

| Method                  | Strength                | Weakness                             |
|-------------------------|-------------------------|--------------------------------------|
| A stick for each object | Never runs out          | Needs as many sticks as objects      |
| Letters a to z as names | Easy to say             | Stops at 26                          |
| Roman symbols           | Short for small numbers | Needs new symbols for larger numbers |

**Example 1.1** A herder keeps one pebble in a bag for each sheep that leaves in the morning. In the evening she removes one pebble for each sheep that returns, and 3 pebbles are left. What does this tell her?

**SOLUTION**

Each pebble is matched with exactly one sheep.

**3 sheep have not returned. She learns this without using any number names.**

## 2 Early ways of counting

- Body parts**: some communities count by pointing to fingers, wrist, elbow, shoulder and so on in a fixed order.
- Tally marks**: one notch for each object. The Lebombo bone (about 44,000 years old, 29 notches) and the Ishango bone (20,000 to 35,000 years old) are the oldest known examples.
- Counting in twos**: the Gumulgal people of Australia used \*urapon\* for 1 and \*ukasar\* for 2, and built other names from them:  $3 = 2 + 1$  is ukasar-urapon,  $4 = 2 + 2$  is ukasar-ukasar,  $5 = 2 + 2 + 1$ ,  $6 = 2 + 2 + 2$ .

- The key idea is **grouping**: count in groups of a fixed size (commonly 2, 5, 10 or 20) and use the group to name larger numbers.
- Most people can tell the size of a collection at a glance only up to about 4. This may be why tally marks are bundled in fives.

**Example 2.1** Write 9 in the Gumulgal style, counting in twos.

SOLUTION

$$9 = 2 + 2 + 2 + 2 + 1$$

**ukasar-ukasar-ukasar-ukasar-urapon**

### 3 Roman numerals

- The Roman system uses several group sizes at once. Numbers that have their own symbol are called **landmark numbers**.

| Symbol | I | V | X  | L  | C   | D   | M    |
|--------|---|---|----|----|-----|-----|------|
| Value  | 1 | 5 | 10 | 50 | 100 | 500 | 1000 |

- To write a number, break it into landmark numbers, taking as many of the largest as possible first.
- A smaller symbol written **before** a larger one is subtracted: IV = 4, IX = 9, XL = 40, XC = 90, CD = 400, CM = 900.
- Weaknesses: there is no zero, larger numbers need new symbols, and multiplication and division are very hard.

**Example 3.1** Write 1947 in Roman numerals.

SOLUTION

$$1947 = 1000 + 900 + 40 + 7$$

M, CM, XL, VII

**MCMXLVII**

**Example 3.2** Add CCXXXII and CCCCXIII without changing them to Hindu numerals.

SOLUTION

Collect the symbols: six Cs, four Xs and five Is.

Five Cs make a D, and five Is make a V.

**DCXLV (that is,  $232 + 413 = 645$ )**

### 4 The idea of a base

- The Egyptians (about 3000 BCE) chose landmark numbers that are all powers of 10: 1, 10, 100, 1000, . . . , with a separate symbol for each. A number was written by repeating each

symbol as often as needed.

- A system whose landmark numbers are  $1, n, n^2, n^3, \dots$  is a **base- $n$  number system**. Base 10 is also called the **decimal** system.
- In a base- $n$  system,  $n$  copies of one landmark number always make the next one. So no symbol is ever needed  $n$  or more times.
- **Why a base helps:** the product of two landmark numbers is again a landmark number ( $n^a \times n^b = n^{a+b}$ ). This makes addition and multiplication systematic. Roman landmark numbers do not have this property.
- Drawback of the Egyptian system: every higher power of 10 needs a brand-new symbol.

**Example 4.1** What are the first five landmark numbers of a base-5 system? Break 143 into these landmark numbers.

**SOLUTION**

$$5^0 = 1, 5^1 = 5, 5^2 = 25, 5^3 = 125, 5^4 = 625$$

$$143 = 125 + 5 + 5 + 5 + 1 + 1 + 1$$

**One 125, no 25s, three 5s and three 1s**

**Example 4.2** In the Egyptian system, how many symbols in all are needed to write 2304?

**SOLUTION**

$$2304 = 2 \times 1000 + 3 \times 100 + 0 \times 10 + 4 \times 1$$

$$2 + 3 + 0 + 4 = 9 \text{ symbols}$$

- The **abacus** used in Europe from about the 11th century was a board with one line for each power of 10. Counters on the lines showed how many of each power were present.

## 5 Place value

- In a **place value system** (positional system) the **position** of a symbol tells which landmark number it counts. No separate symbols for the landmark numbers are needed, so a few symbols can write every number.

| System                    | Base                            | Notes                                                                                                     |
|---------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------|
| Mesopotamian (Babylonian) | 60                              | Symbols for 1 and 10 made the numbers 1 to 59; blank space for an empty place, later a placeholder symbol |
| Mayan                     | 20, with places 1, 20, 360, ... | Dot for 1, bar for 5, a shell symbol for zero; written top to bottom                                      |
| Chinese rod numerals      | 10                              | Rod patterns for 1 to 9, alternating upright and sideways forms; blank for an empty place                 |
| Hindu (Indian)            | 10                              | Ten digits 0 to 9, with 0 a full digit and a number                                                       |

- The base-60 system survives in our units of time: 1 hour = 60 minutes and 1 minute = 60 seconds.

- Without a symbol for an empty place, a numeral can be read in more than one way. A **placeholder (zero)** removes this confusion.

**Example 5.1** Break 7530 into Mesopotamian places.

SOLUTION

$$7530 = 2 \times 3600 + 5 \times 60 + 30$$

The three places hold 2, 5 and 30

**Example 5.2** Break 500 into Mayan places (1, 20, 360).

SOLUTION

$$500 = 1 \times 360 + 7 \times 20 + 0 \times 1$$

The places hold 1, 7 and 0, written from top to bottom

## 6 The Hindu number system

- It is a base-10 place value system with ten digits 0, 1, 2, ..., 9. For example  $375 = 3 \times 10^2 + 7 \times 10 + 5 \times 1$ .
- It began in India about 2000 years ago. The Bakhshali manuscript (about the 3rd century CE) writes zero as a dot. Āryabhaṭa (499 CE) computed extensively with the system.
- Brahmagupta (628 CE) gave rules for arithmetic with **zero as a number** and with negative numbers.
- It reached the Arab world by about 800 CE through Al-Khwārizmī and Al-Kindi, and Europe by about 1100 CE. Fibonacci argued for it around 1200, and it was in general use in Europe by the 17th century.
- Europeans learned the numerals from Arab scholars and called them "Arabic numerals". Arab scholars themselves called them Hindu numerals. Today they are called Hindu, Indian or Hindu-Arabic numerals.
- Because zero is a digit and each place holds exactly one digit, every number can be written in exactly one clear way, and calculation is easy.

| Stage | Idea                                                       |
|-------|------------------------------------------------------------|
| 1     | Count in groups of one size                                |
| 2     | Group using landmark numbers                               |
| 3     | Make the landmark numbers the powers of one number: a base |
| 4     | Let position show the landmark number: place value         |
| 5     | Use 0 as a digit and as a number                           |

## 7 Writing numbers in other bases

- The same place value idea works for any base  $n$ , using the digits 0 to  $n - 1$ .

- To convert a number to base  $n$ : take out the largest power of  $n$  as many times as possible, then the next power, and so on. The counts are the digits.
- To convert back: multiply each digit by its power of  $n$  and add.

**Example 7.1** Write 25 in base 8, base 5 and base 2.

**SOLUTION**

Base 8:  $25 = 3 \times 8 + 1$ , giving 31.

Base 5:  $25 = 1 \times 25 + 0 \times 5 + 0$ , giving 100.

Base 2:  $25 = 16 + 8 + 1 = 1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1$ , giving 11001.

**31 (base 8), 100 (base 5), 11001 (base 2)**

**Example 7.2** What number does the base-5 numeral 2133 stand for?

**SOLUTION**

$$2 \times 125 + 1 \times 25 + 3 \times 5 + 3 \times 1$$

**293**

## 8 Common mistakes

- Writing 4 as IIII or 40 as XXXX instead of IV and XL, or reading IX as 11.
- Mixing up a base with a group size. A base needs **every** landmark number to be a power of the same number.
- Thinking the Egyptian system had place value. It had a base but used separate symbols.
- Using the digit  $n$  in base  $n$ . Base 5 uses only 0, 1, 2, 3, 4.
- Dropping a zero in the middle of a numeral: 3605 and 365 are different numbers.